MATTHEW LONGORIA, EIT

Civil Design Engineer | Fiber Network Design | GIS
Arlington, TX | mattalongoria@gmail.com | 817-713-5285
Portfolio: fieldandsignal.tech (Work in progress)

PROFESSIONAL SUMMARY - GIS professional and Texas EIT supporting electric transmission, telecommunications, and water resources projects. Experience with Esri spatial data maintenance, data quality review, ArcGIS Online templates, GIS troubleshooting, and cartography. Combines infrastructure design experience with a master's in civil engineering and a graduate GIS certificate.

CORE COMPETENCIES

- Fiber Backbone & FTTH Design
- OSP Route Design
- Vetro Fiber / Katapult
- ArcGIS Pro / ArcGIS Online
- Spatial Data QA/QC
- Utility Infrastructure Mapping
- CAD Design & Record Drawings
- Permitting Coordination
- Remote Project Delivery

PROFESSIONAL EXPERIENCE

Aquila Corporation | OSP Design Engineer — Remote | Nov 2024 –

- Designed and drafted 200+ miles of backbone and FTTH infrastructure and 100+ miles of backbone for GLO Fiber Ohio
- Performed fiber design in Vetro Fiber and route design in Katapult on projects exceeding 100,000 feet of path
- Created ArcGIS Online programming templates and supported ArcGIS troubleshooting for design-team workflows
- Coordinated with permitting authorities, field crews, and project managers across multiple jurisdictions

FSC Engineers | EIT — Remote | Nov 2023 – Oct 2024

- Designed and engineered XTO Lease Drilling pads using Bentley OpenRoads including grading, drainage, and site civil design
- Completed Plan and Profile Sheets for subdivision civil design projects, Supported remote civil design workflows in a fast-paced production environment

Patterson & Dewar Engineers | Civil Engineering Associate — Remote | May 2023 – Nov 2023

- Prepared SPCC Plans for electric cooperative clients across South Carolina, Alabama, and Tennessee addressing oil spill prevention
- Completed site civil plans for substation and transmission line construction projects
- Supported GIS department with spatial database management, analysis, and cartography
- Prepared Storm Water Pollution Prevention Plans (SWPPP) for transmission line projects in North Carolina

Collabra / Oncor | GIS Analyst — Contract Hybrid Fort Worth | May 2022 – May 2023

- Maintained and updated Oncor Transmission GIS system for electrical transmission spatial data using ESRI products
- Performed QA/QC on GIS data from new build plans, tested system upgrades, and managed spatial database accuracy
- Facilitated acquisition and import of external GIS data sources into Oncor GIS datasets
- Managed easements, database records, and cartographic products for transmission infrastructure

CCI Systems | Design Engineer — Remote | 2021 – 2022

- Designed site plans for fiber and coaxial cable route updates using CAD and GIS platforms, updated existing designs and maintained project drawing revisions and record drawings

HDR | *Civil* EIT — Dallas TX | Jan 2019 – Feb 2020

- Completed H&H modeling and stormwater systems modeling for drainage infrastructure projects
- Performed FEMA flood mapping using ESRI GIS products for regulatory compliance
- Prepared cost estimates and technical reports for TxDOT drainage, BNSF Railroad drainage, and NRCS dam rehabilitation projects
- Supported FEMA floodplain studies and dam rehabilitation design under senior engineer supervision

TECHNICAL SKILLS

GIS Platforms	ArcGIS Pro, ArcMap, ArcGIS Online, QGIS, Spatial NET, Google Earth Pro
Telecom / OSP	Vetro Fiber Design, Katapult Engineering, LODE data, backbone / FTTH design
CAD / Drafting	Bentley MicroStation, Bentley OpenRoads, Autodesk Civil 3D, Bluebeam, MS Visio
Programming	ArcPy, Java, ArcGIS Online automation

EDUCATION & CERTIFICATIONS

Engineer in Training (EIT) | Texas Board of Professional Engineers | License #67963
GIS Graduate Certificate | University of Texas at Arlington | GPA 4.0 | Urban Planning Focus
Master of Engineering, Civil Engineering | Texas A&M University | GPA 3.2 | Water Resources Focus
Bachelor of Science, Geophysics | Texas A&M University | GPA 3.1 | Minors: Geology, Math